

Income and Crime Rate Association with Housing Prices Across California Counties

Final Project Report

Author: Samira Bayat Kazimi

Github link: <https://github.com/skazimi/california-county-housing-income-crime>

1. Motivation and Rationale

Housing affordability is one of the most important social and economic issues in California. California is often called the Golden State, but it is also known as one of the most expensive places to live in the United States. Housing prices are very high in many counties, while income levels and public safety conditions can be very different from one county to another. Because of this, looking only at housing prices does not give a complete picture of living conditions across California.

This project combines housing values, median household income, and violent crime rates into one integrated county level dataset. By bringing these data sources together, counties can be compared from several perspectives instead of only one.

The main research question is:

What factors are associated with housing prices across California counties?

More specifically, this project examines two questions:

1. Do counties with higher median household income tend to have higher housing prices?
2. Are higher violent crime rates associated with lower housing prices?

These questions are interesting because housing values may be related to economic conditions, but other community factors, such as public safety, may also matter when evaluating local living conditions. To answer these questions, the project uses descriptive statistics and visualizations such as scatter plots, bar charts, comparison graphs, and correlation analysis. These methods help show how economic and public safety factors may be connected to housing patterns across California counties.

This project is descriptive and exploratory. It does not claim that income or crime directly causes housing prices to increase or decrease. Instead, it examines patterns and relationships in the integrated county-level data.

In addition, this project demonstrates a complete data workflow. It begins with data collection from web and API sources, continues with data cleaning and preprocessing, integrates multiple datasets using a shared county level key, and ends with analysis and visualization. The final result is a reproducible project that can be rerun, reviewed, and extended in the future.

2. Description of Data Sources

This project uses three public data sources to study county-level relationships among housing values, median household income, and violent crime rates in California. The data sources are Zillow housing data, American Community Survey income data from the U.S. Census API, and California violent crime data from California open data. Each dataset was collected programmatically, cleaned into a consistent format, and prepared for integration by county name.

2.1 Zillow Housing Data

The first data source is **Zillow housing data**. This project uses county level Zillow Home Value Index (ZHVI) data. ZHVI is used as the housing value measure because it provides an estimated typical home value for each county.

The data collection process begins from the Zillow Research Data page: <https://www.zillow.com/research/data/>

The **src/scrapper.py** script searches this webpage for a usable county level ZHVI csv file containing

"County_zhvi" in the link. If the scraper does not find a matching file, it uses a backup Zillow csv link:

https://files.zillowstatic.com/research/public_csvs/zhvi/County_zhvi_uc_sfrcondo_tier_0.33_0.67_sm_sa_month.csv

After downloading the file, the scraper filters the rows to keep only California counties, identifies the **latest available monthly ZHVI** column, and creates a smaller county level table. The **src/get_data.py** script saves both the full Zillow download and the filtered California Zillow file in the **data/raw** folder.

During preprocessing, **src/clean_data.py** standardizes county names, removes extra spaces, keeps only rows where the state is California, removes rows with empty county names or ZHVI values, and keeps only the columns needed for later analysis **county_name**, **state**, and **zhvi**.

After filtering, the Zillow dataset contains **58** California county level samples.

2.2 ACS Income Data

The second data source is **American Community Survey (ACS)** data from the U.S. Census API. This project uses median household income as the income measure. The ACS data provides economic context for comparing housing values across California counties.

The project collects 2024 ACS 5 year county level data for California using this API request:

https://api.census.gov/data/2024/acs/acs5?get=NAME,B19013_001E&for=county:*&in=state:06

In this request:

- **NAME** provides the county name.
- **B19013_001E** provides the median household income estimate.
- **county:*** requests all counties.
- **state:06** limits the results to California.

The **src/get_data.py** script sends the API request and saves the raw response as two formats:

- **acs_income_raw.json**
- **acs_income_raw.csv**

Saving both formats keeps the original data and also provides a csv file that is easier to clean later.

During preprocessing, **src/clean_data.py** renames the Census income field **B19013_001E** to **median_income**, formats the state code, extracts the county name from the original Census NAME field, standardizes county names, keeps only California rows, removes rows with missing income values, and keeps only the fields needed for integration:

- **county_name**
- **median_income**

The ACS API key is not written directly inside the code for privacy. Instead, the project reads it from an environment variable named **ACS_API_KEY**. A valid ACS API key is required before running the data download script. The ACS dataset contains **58** California county level samples.

2.3 California Violent Crime Data

The third data source is **California violent crime** data from California open data. This dataset provides county level violent crime rate information across multiple years.

The crime data is collected from this csv file:

<https://data.ca.gov/datastore/dump/358f23e6-f0f9-45e1-864c-fbad0f0b84b3?bom=True>

The **src/get_data.py** script downloads the raw crime csv file and saves it as: **ca_violent_crime_raw.csv**

Because the raw crime dataset contains records from multiple years, the preprocessing step in **src/clean_data.py** prepares it for county level integration. The script standardizes county names, removes unusable rows, converts **reportyear** to numeric format, sorts records by year, and keeps only the latest available record for each county. It also renames rate column to **violent_crime_rate**.

The final cleaned fields are:

- **county_name**
- **violent_crime_rate**
- **reportyear**

After keeping only the latest year for each county, the cleaned crime dataset contains 58 California county-level samples. The latest county level violent crime rate available in this dataset is from 2013. Therefore, the analysis uses this crime measure as a historical county level indicator rather than a fully current measure, while housing and income data are more recent. This time difference should be considered when interpreting the results.

After the three cleaned datasets are merged by **county_name**, the final integrated dataset is saved as **data/processed/integrated_county_data.csv**

It contains county level variables for housing value, median household income, and violent crime rate which are later used for analysis and visualization.

Source	Purpose	Main Fields	Project File
Zillow Housing Data	Housing values by county	county_name, state, zhvi	scraper.py get_data.py clean_data.py
ACS Census API	Median household income by county	county_name, median_income	get_data.py clean_data.py
California Crime Data	Violent crime rate by county	county_name, violent_crime_rate, reportyear	get_data.py clean_data.py

2.4 Changes from the Original Plan and Challenges Resolved

The overall project goal stayed the same as the original plan combining housing, income, and crime data to compare county level patterns in California. However, some implementation details changed during the project, and several challenges were resolved to make the data collection, cleaning, and integration process more reliable.

One change was the Zillow data collection process. Instead of depending only on one fixed download link, the final project uses a scraper to search the Zillow Research Data page for a usable county level ZHVI csv file. A backup Zillow csv link was also added in case the scraper does not find the expected file. This made the Zillow collection process more reliable.

Another challenge was that the three datasets used different county name formats and structures. Since **county_name** is the common key used for merging the datasets, county names had to be standardized before integration. This step helped reduce mismatches among the Zillow, ACS, and crime data.

Another change from the original plan was the tool used for data cleaning. At first, I planned to work with the built in csv module. After learning pandas in class, I decided to use pandas instead. Pandas made the work easier because it allowed me to read, filter, clean, and save the datasets more efficiently. This change also made the code shorter and easier to organize.

A further challenge was that the crime dataset contained multiple years of data for each county, while the final integrated dataset needed only one county level crime value per county. To resolve this, the cleaning script sorts the crime records by **reportyear** and keeps only the latest available record for each county.

The project also handled the ACS API key carefully. Instead of writing the key directly in the code, the script reads it from an environment variable. This protects private information and makes the project safer to share publicly.

3. Integrated Data Model

The integration step is performed by src/integrate_data.py. The cleaned Zillow, ACS, and crime datasets are joined using county_name. The final integrated file is saved as: data/processed/integrated_county_data.csv. The integrated dataset contains the combined variables:

- county_name
- zhvi
- median_income
- violent_crime_rate
- reportyear

The integration uses outer joins, so it keeps all counties that appear in the cleaned datasets, even if some counties are missing from one of the sources. After merging the data, the script keeps only the five columns needed for analysis.

The cleaned datasets are prepared to support a county level merge. Zillow provides California county housing value records, ACS provides county level median income records, and the crime dataset is reduced to the latest available record for each county.

After cleaning, each input table is organized at the county level, with one record per county. Therefore, the intended relationship among the cleaned Zillow, ACS, and crime tables is one-to-one by county_name. This makes it possible to combine the three sources into one final county level analysis table.

The diagram shows that `county_name` is the common key connecting the Zillow clean table, ACS income clean table, and crime clean table. The final integrated county dataset contains the combined columns `county_name`, `zhvi`, `median_income`, `violent_crime_rate`, and `reportyear`. This model explains how the combined data system works and why standardizing county names was necessary before merging the datasets.

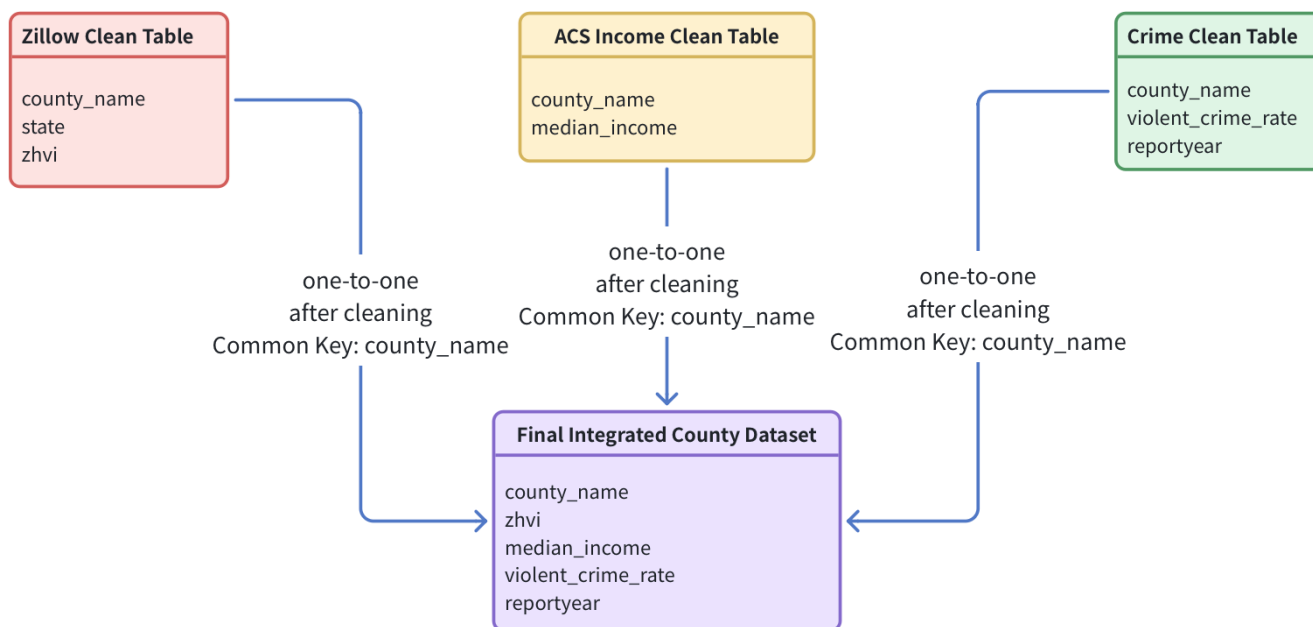


Figure 1. Integrated data model diagram showing how the cleaned datasets connect by `county_name`.

4. Analyses and Visualizations

The analysis and visualization step is performed by `src/analyze_visualize.py`. This script reads the final integrated county dataset, converts the main analysis variables to numeric values, removes rows with missing values in those variables, saves summary statistics, and creates several visualizations.

The main analysis techniques used in this project are descriptive summary statistics, correlation analysis, ranking counties by housing value and violent crime rate, and visual comparison using scatter plots and bar charts. The script saves summary statistics for the three numeric variables, calculates correlations between them, identifies the top ten counties by housing value and crime rate, and creates figures to show the patterns more clearly.

The analysis focuses on three main variables:

- Zillow Home Value Index (`zhvi`)
- Median household income (`median_income`)
- Violent crime rate (`violent_crime_rate`)

The visualizations are used to compare these variables across California counties and to identify patterns that may not be obvious from the table alone.

4.1 Correlation Analysis

The correlation heatmap shows the linear relationships among housing value, median household income, and violent crime rate. This figure was created by calculating a correlation matrix for `zhvi`, `median_income`, and `violent_crime_rate`, and then displaying the values as a heatmap. Based on the generated correlation figure, the strongest relationship appears between Zillow Home Value Index and median income. This suggests that counties with higher median household income also tend to have higher housing values in the integrated dataset.

In contrast, violent crime rate shows a much weaker linear relationship with housing value and median income. This means that, in this county level dataset, crime rate does not appear to be strongly associated with the other two variables through a simple linear correlation.

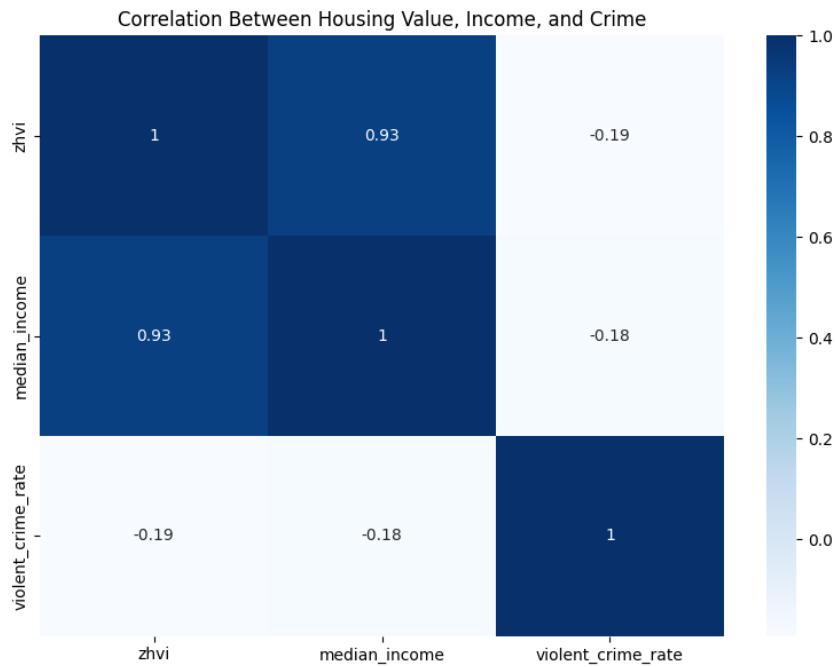


Figure 2. Correlation between Zillow Home Value Index, median income, and violent crime rate.

4.2 Top Counties by Housing Value

The top housing value bar chart ranks the ten counties with the highest Zillow Home Value Index in the final analysis dataset. This chart was made by sorting counties from highest to lowest zhvi value and selecting the top ten counties. This figure helps show how housing values differ across California counties and highlights the counties with the most expensive estimated home values.

The chart can help support the broader comparison between county-level housing values and other conditions such as income and crime rate.

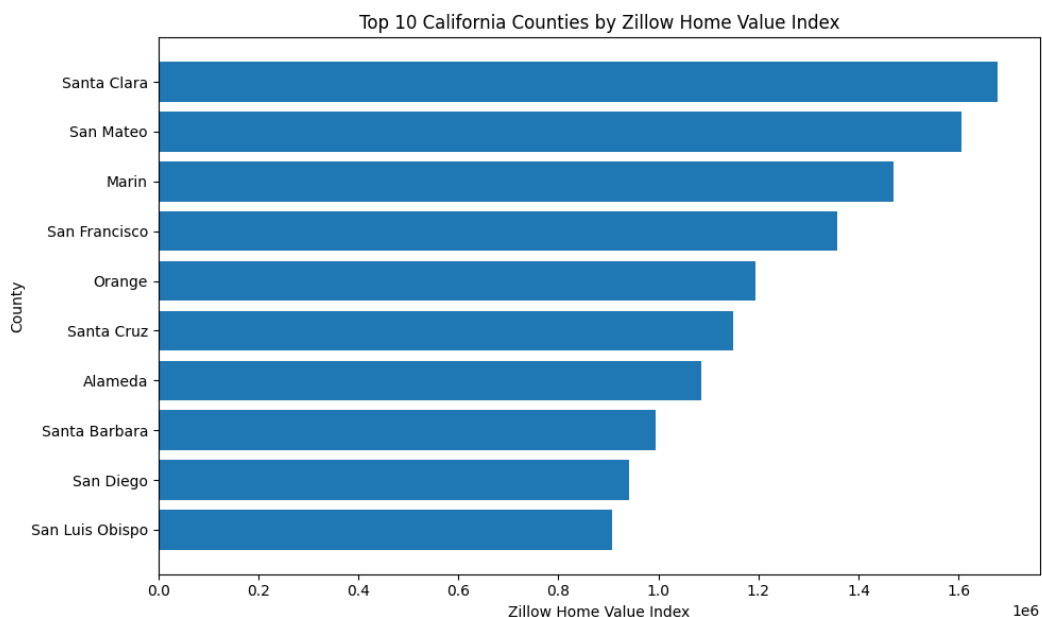


Figure 3. Top 10 California counties by Zillow Home Value Index.

4.3 Top Counties by Violent Crime Rate

The violent crime rate bar chart ranks the ten counties with the highest violent crime rates in the final analysis dataset. This chart was made by sorting counties from highest to lowest violent_crime_rate and selecting the top ten counties. This figure shows that violent crime rates are not evenly distributed across counties. Some counties have much higher reported violent crime rates than others.

This makes crime rate an important variable to compare with housing values and income, even if it does not show a strong linear correlation in the overall analysis.

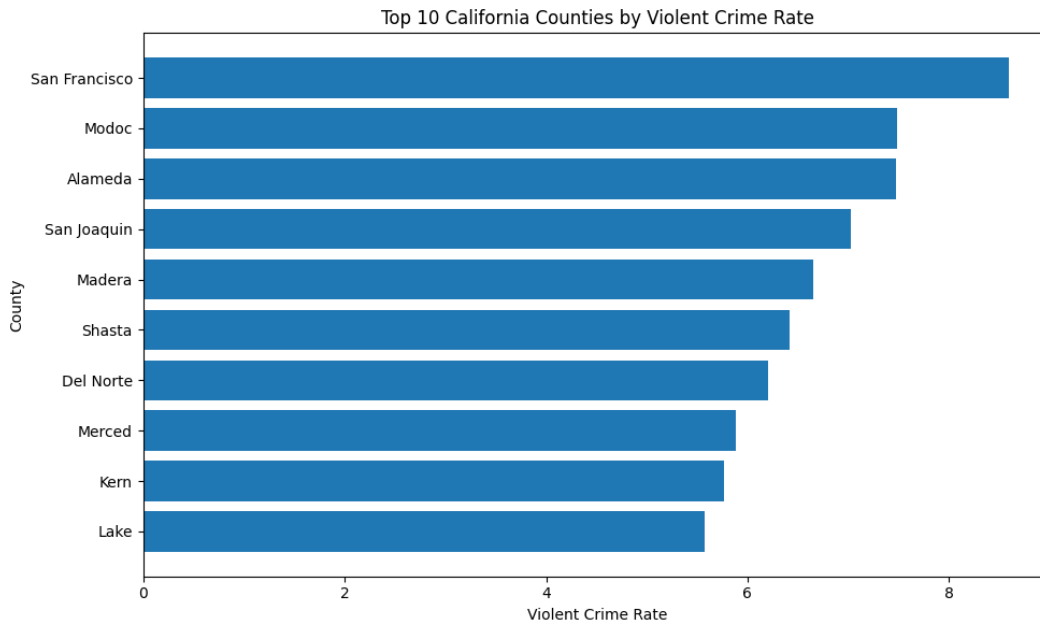


Figure 4. Top 10 California counties by violent crime rate.

4.4 Median Income Compared with Housing Value

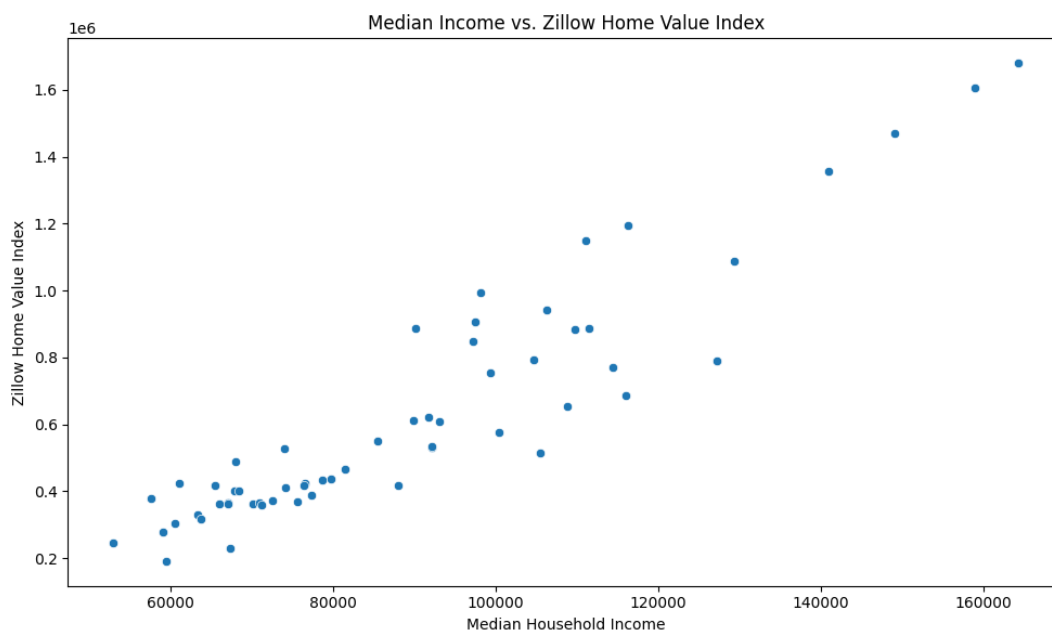


Figure 5. Median household income compared with Zillow Home Value Index.

The scatter plot comparing median household income and Zillow Home Value Index shows the relationship between county income levels and housing values. The x-axis shows median household income, and the y-axis shows Zillow Home Value Index. The upward pattern in the figure suggests that counties with higher median household income generally also have higher housing values. This scatter plot visually supports the positive relationship shown in the correlation heatmap.

4.5 Violent Crime Rate Compared with Housing Value

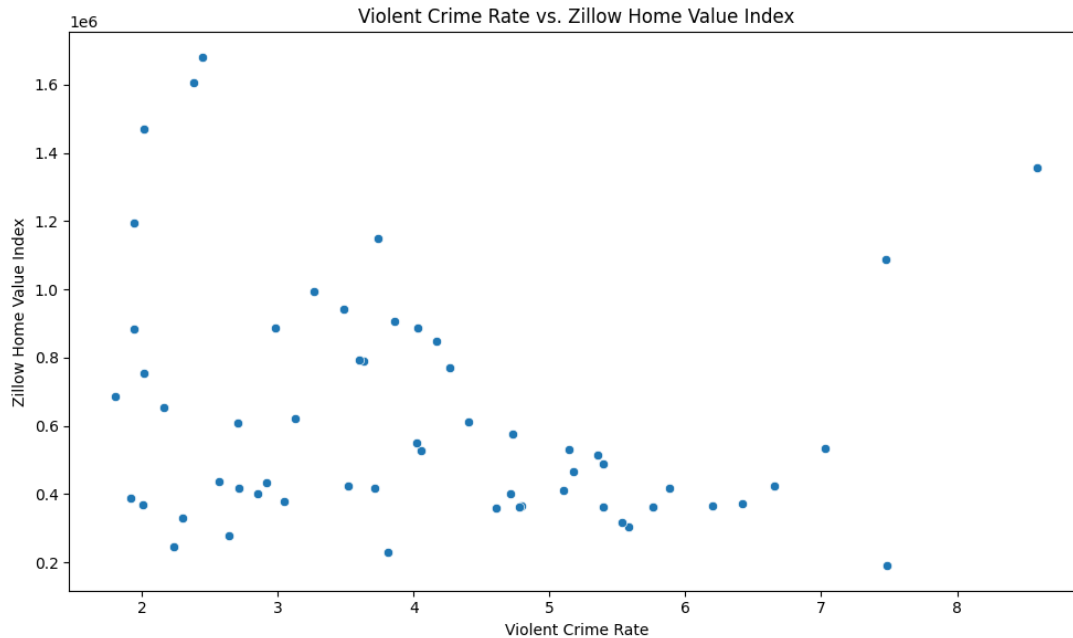


Figure 6. Violent crime rate compared with Zillow Home Value Index.

The scatter plot comparing violent crime rate and Zillow Home Value Index is used to examine whether counties with higher crime rates tend to have lower housing values. The x-axis shows violent crime rate, and the y-axis shows Zillow Home Value Index. In the figure, the points do not form a clear upward or downward pattern.

This suggests that violent crime rate does not show a strong simple linear relationship with Zillow Home Value Index in the final county level dataset.

5. Conclusions and Findings

The main conclusion of this project is that integrating multiple county level datasets creates a more useful view of California counties than analyzing one dataset alone. Housing value, income, and crime represent different parts of county conditions. By joining the datasets together, the project makes it possible to compare these variables side by side.

The strongest finding is the positive relationship between median household income and Zillow Home Value Index. Counties with higher income generally show higher housing values. This pattern is visible in both the income-versus-housing scatter plot and the correlation heatmap.

In contrast, the relationship between violent crime rate and housing value is much weaker. Violent crime rates vary across counties, but the scatter plot and correlation heatmap do not show a strong simple linear connection between crime rate and housing value. These results suggest that, in this descriptive county level analysis, housing value is more closely connected to income rather than to violent crime rate.

The project also shows the importance of cleaning and standardization. The datasets came from different public sources and used in different formats. Without cleaning county names and preparing consistent fields, the datasets could not be reliably merged into one integrated analysis file.

These findings are useful because they show how combining public datasets can help compare local economic and community conditions, and they suggest that income may be a more informative factor than violent crime rate when examining county level housing value patterns in this dataset.

The conclusions are descriptive and the analysis does not prove that higher income causes housing values to increase or that crime directly changes housing prices in opposite direction. Instead, the results show patterns in the available county level data and provide a foundation for future analysis.

6. Future Work

This project could be extended by adding more county level data sources. Since the current integration key is `county_name`, a new dataset could be joined if it includes a compatible county field. Possible future sources include education level, unemployment rate, poverty rate, population size, school quality, healthcare access, or transportation access. These additional variables would make the analysis more complete and could help explain housing values from multiple social and economic perspectives.

Another future direction is to expand the project from California counties to more detailed geographic levels, such as ZIP codes or cities. ZIP code level analysis would allow more precise comparisons between housing prices, income, crime, education, and other local conditions. Counties can be broad, so smaller geographic units may better capture neighborhood level differences.

The project could also be expanded beyond California. In the future, I could extend the analysis to the entire United States or study each state separately using the same county level framework. This would make it possible to compare broader housing, income, and crime patterns across different regions.

A major future improvement would be creating an interactive application. In this app, users could search by ZIP code, county, city, or state and view housing values together with crime rate, income, education level, and other local factors. This would make the project more practical, easier to explore, and more useful for people who are not familiar with Python or data analysis tools.

Finally, a future improvement would be to replace the historical crime dataset with a more recent county-level crime source, if a similarly structured and accessible dataset becomes available.